

ConvNet Size & Parameters

Advanced · ANSWER KEY

A color image through a deeper ConvNet — no number repeats, so lean on the formulas.

THE ONLY TWO FORMULAS (stride 1, no padding)

Conv2D: $\text{params} = ((M \times N \times L) + B) \times F$ $\text{new size} = \text{old size} - (M - 1)$
 MaxPool: $\text{params} = 0$ $\text{new size} = \text{old size} / \text{pool}$
 Flatten: $\text{params} = 0$ $\text{size} = \text{rows} \times \text{cols} \times \text{channels}$
 Dense: $\text{params} = (\text{inputs} + 1) \times \text{nodes}$ $\text{size} = \text{nodes}$

M, N = kernel size L = channels coming IN (inherited, never chosen)

B = bias = 1 F = filters = feature maps coming OUT (you choose)

THE CODE (given)

```
model = models.Sequential()
model.add(layers.Conv2D(16, (5, 5), activation='relu',
                        input_shape=(64, 64, 3)))
model.add(layers.MaxPooling2D((2, 2)))
model.add(layers.Conv2D(32, (3, 3), activation='relu'))
model.add(layers.MaxPooling2D((2, 2)))
model.add(layers.Conv2D(64, (3, 3), activation='relu'))
model.add(layers.MaxPooling2D((2, 2)))
model.add(layers.Flatten())
model.add(layers.Dense(128, activation='relu'))
model.add(layers.Dense(1, activation='sigmoid'))
```

Layer	Output shape	Params	The math
Conv2D: F=16, (5x5)	(60, 60, 16)	1,216	$64 - (5 - 1) = 60$; $((5 \times 5 \times 3) + 1) \times 16$ L=3 (RGB)
MaxPool 2x2	(30, 30, 16)	0	$60 / 2 = 30$
Conv2D: F=32, (3x3)	(28, 28, 32)	4,640	$30 - (3 - 1) = 28$; $((3 \times 3 \times 16) + 1) \times 32$ L=16 in
MaxPool 2x2	(14, 14, 32)	0	$28 / 2 = 14$
Conv2D: F=64, (3x3)	(12, 12, 64)	18,496	$14 - (3 - 1) = 12$; $((3 \times 3 \times 32) + 1) \times 64$ L=32 in
MaxPool 2x2	(6, 6, 64)	0	$12 / 2 = 6$
Flatten	(2304,)	0	$6 \times 6 \times 64 = 2304$
Dense(128)	(128,)	295,040	$(2304 + 1) \times 128$
Dense(1, sigmoid)	(1,)	129	$(128 + 1) \times 1$
TOTAL trainable parameters		319,521	

Answer key — every shape and count worked out.